# CAHIER DES CHARGES

*Specifications Document*

# FLIPPER 42

*Portable Multi-Protocol Security Research Device*

| | |
|---|---|
| **Project** | FLIPPER 42 |
| **School** | École 42 Paris |
| **Module** | Electronics / Embedded Systems |
| **Document type** | Cahier des Charges (CDC) |
| **Version** | 1.0 |
| **Date** | June 2026 |
| **Status** | Draft - for academic validation |
| **Authors** | FLIPPER 42 Team - 42 Paris |

# Table of Contents

# 1. Introduction

## 1.1 Project Overview

The FLIPPER 42 is a portable, open-hardware multifunctional device designed for interaction with various electronic access-control and radio-frequency systems. Inspired by the Flipper Zero, the project explores low-level embedded systems design and RF security research using an alternative hardware architecture built around the ATmega2560 microcontroller.

The device is conceived for:

- Educational purposes and academic training in embedded systems
- Security research and ethical hacking activities
- Reading, cloning, and replaying NFC/RFID credentials on authorised systems
- Sub-GHz radio signal acquisition and controlled reproduction
- Hands-on exploration of hardware security protocols

## 1.2 Scope

This document describes the technical and functional specifications of the FLIPPER 42 hardware platform, covering architecture, communication modules, power supply, firmware behaviour, and acceptance criteria. It serves as the reference baseline for the 42 Paris electronics module validation.

## 1.3 Definitions & Abbreviations

| Acronym / Term | Definition |
|---|---|
| CDC | Cahier des Charges |
| MCU | Microcontroller Unit |
| RFID | Radio-Frequency IDentification |
| NFC | Near-Field Communication |
| Sub-GHz | Radio frequencies below 1 GHz (e.g. 433 MHz, 868 MHz) |
| SPI | Serial Peripheral Interface - synchronous serial bus |
| I2C | Inter-Integrated Circuit - two-wire serial bus |
| UART | Universal Asynchronous Receiver-Transmitter |
| BMS | Battery Management System |
| ADC | Analog-to-Digital Converter |
| GPIO | General-Purpose Input/Output |
| EEPROM | Electrically Erasable Programmable Read-Only Memory |
| LF | Low Frequency (125 kHz RFID band) |
| HF | High Frequency (13.56 MHz NFC band) |

# 2. Functional Requirements

## 2.1 Core Functions

The device shall implement the following primary functions:

1. RF Signal Detection & Identification - detect and classify incoming radio signals in supported bands
2. NFC Credential Reading - read and decode ISO 14443 and ISO 15693 compliant tags at 13.56 MHz
3. RFID Credential Reading - read 125 kHz LF tags using the EM4095 transceiver
4. Data Storage - persist captured payloads, signal recordings, and configurations on microSD
5. Signal Replay - reproduce previously recorded signals upon authorised user action
6. External Expansion - expose hardware interfaces (SPI, UART, GPIO) for optional add-on modules

## 2.2 Sub-GHz (CC1101)

## 2.3 NFC (ST25R3916)

The ST25R3916B-AQWT high-performance NFC frontend shall support:

- Reader/Writer mode for ISO 14443-A/B and ISO 15693 tagsAlso @
- Card emulation mode (Type A / Type B)
- Reading, storing, and replaying NDEF messages and raw tag dumps
- Communication with the MCU via SPI at 3.3 V

## 2.4 RFID (EM4095)

The EM4095 125 kHz RFID transceiver shall:

- Read EM4100-compatible and compatible LF tags
- Provide demodulated data via GPIO/UART to the MCU
- Store captured UIDs and data payloads to microSD

## 2.5 User Interface

The device shall provide a 6-button navigation interface:

| Button | Direction | Function |
|---|---|---|
| ◄ | Left | Navigate left / back |
| ▼ | Down | Scroll down / decrease value |
| ● | Select | Confirm / enter |
| ▲ | Up | Scroll up / increase value |
| ► | Right | Navigate right / forward |
| ⮐ | Outside / Isolated | Back / Return |

An OLED display (connected via I2C) shall provide a menu-driven interface showing:

- Main menu with module selection
- Real-time signal activity and status
- Stored file browser for saved captures
- Battery level and system status

# 3. Hardware Architecture

## 3.1 System Block Diagram Overview

The hardware architecture is centred on the ATmega2560 MCU and interconnects the following subsystems via dedicated bus protocols as described in the uploaded block diagram:

| Subsystem | Interface | Voltage | Notes |
|---|---|---|---|
| MCU - ATmega2560 | Central | 5 V | Main controller, 256 KB Flash, 8 KB SRAM, 4 KB EEPROM |
| OLED Display | I2C | 3.3/5V | ER-OLED015-2Y, 128×64 px |
| microSD Card | SPI | 3.3 V | FAT32 formatted; stores captures & config |
| CC1101 Sub-GHz Transceiver | SPI | 3.3 V | Level-shifted; 433 MHz only; external antenna |
| ST25R3916 NFC Frontend | SPI | 3.3 V | ISO 14443/15693 reader + card emulation |
| EM4095 RFID Transceiver | GPIO | 5 V | 125 kHz LF RFID |
| Battery + BMS | Power | 3.7 V | LiPo cell with protection circuit |
| USB / UART Bridge | UART | 5 V | PC connectivity, firmware upload |
| Buttons (5×) | GPIO | 3.3 V | Navigation; debounced in firmware |
| Level Shifter | Logic | 3.3/5V | Bidirectional; protects 3.3 V peripherals |
| Voltage Divider / ADC | Analog | Var. | Battery voltage sense via MCU ADC |

## 3.2 Microcontroller - ATmega2560

The ATmega2560 (TQFP-100) is the central processing unit. Key specifications:

| Parameter | Value |
|---|---|
| Architecture | 8-bit AVR RISC |
| Clock speed | Up to 16 MHz (external crystal) |
| Flash memory | 256 KB (in-system programmable) |
| SRAM | 8 KB |
| EEPROM | 4 KB |
| GPIO | 86 pins |

| | |
|---|---|
| SPI interfaces | 1× hardware SPI (shared with CC1101, SD, NFC) |
| I2C (TWI) | 1× hardware TWI (OLED) |
| UART | 4× hardware UART (1 used for USB bridge) |
| ADC | 16-channel, 10-bit |
| Timers | 6× (8-bit and 16-bit) |
| Operating voltage | 4.5 V – 5.5 V |

## 3.3 Communication Buses

### 3.3.1 SPI Bus (3.3 V)

A single SPI bus (MOSI, MISO, SCK) serves the CC1101, microSD, and ST25R3916. Each peripheral is independently selected via a dedicated active-low CS (chip select) line driven by the MCU GPIO. The bus operates at 3.3 V logic; a bidirectional level shifter protects all 3.3 V devices from the 5 V MCU outputs.

### 3.3.2 I2C Bus

The I2C bus connects the OLED display to the MCU using standard 400 kHz fast-mode. Pull-up resistors (4.7 kΩ typical) shall be fitted on SDA and SCL lines.

### 3.3.3 UART

UART0 is routed to a USB-to-UART bridge IC (e.g. CH340G or CP2102) for PC connectivity. This enables firmware flashing via avrdude and runtime serial debugging at 115200 baud.

## 3.4 Power Supply

| Domain | Source | Regulation | Consumers |
|---|---|---|---|
| 5 V main | USB or LiPo via boost | LDO / boost converter | MCU, EM4095, USB bridge |
| 3.3 V rail | AMS1117-3.3 or equiv. | LDO from 5 V | CC1101, NFC, OLED, SD, level shifter |
| Battery | LiPo 3.7 V / 1000 mAh+ | BMS + protection IC | System supply when off USB |

The BMS shall provide over-charge, over-discharge, and short-circuit protection. A voltage divider on the battery line feeds the MCU ADC for state-of-charge estimation.

# 4. Non-Functional Requirements

## 4.1 Physical

| Requirement | Target |
|---|---|
| Weight | TODO < 400g |
| Enclosure | PCB-mount shell Initially then 3D-printed |
| Connector | USB-C or Micro-USB for power and programming |

## 4.2 Electrical

| Requirement | Target |
|---|---|
| Supply input | 5 V via USB; 3.7 V LiPo battery |
| 3.3 V rail | +-5% tolerance |
| Autonomy | TODO |

## 4.3 Software & Firmware

- Firmware written in C using the “avr” library toolchain
- Menu-driven OLED interface with 5-button navigation
- FAT32 filesystem on microSD
- UART debug console at 115200 baud accessible via USB

## 4.4 Safety

The FLIPPER 42 shall be operated exclusively in compliance with applicable radio regulations.

The device is intended for educational and security research only.

# 5. Bill of Materials (BoM)

The following table lists the principal components identified for the FLIPPER 42 prototype. Quantities are for a single unit.

| Qty | Component | Part Number | Interface | Notes |
|---|---|---|---|---|
| 1 | MCU | ATmega2560-16AU | SPI/I2C/UART/GPIO | Main controller, TQFP-100 |
| 1 | Sub-GHz Transceiver | CC1101 | SPI 3.3 V | 433 MHz; external antenna |
| 1 | NFC Frontend | ST25R3916B-AQWT | SPI 3.3 V | ISO 14443 / 15693 |
| 1 | RFID Transceiver | EM4095 | GPIO 5 V | 125 kHz LF RFID |
| 1 | OLED Display | ER-OLED015-2 | I2C | 128×64 px |

| | | | | |
|---|---|---|---|---|
| 1 | microSD Socket | 1040310811 (Molex) | SPI 3.3 V | FAT32; micro-SD |
| 1 | 3.3 V LDO | AMS1117-3.3 | Power | 300 mA min. |
| 1 | Battery BMS | | Power | LiPo protection |
| 1 | LiPo Battery | 3.7 V / 1000+ mAh | Power | Rechargeable |
| 1 | Coaxial Antenna | 433 MHz SMA | RF | External removable |
| 5 | Tactile Switches | Generic 6×6 mm | GPIO | Navigation buttons |
| -- | Passives (R, C, L) | Various | PCB | Per datasheet values |

# 6. Implementation Procedure

## 6.1 Development Phases

| Phase | Activity | Deliverable |
|---|---|---|
| 1 - Design | Schematic capture, component selection, BOM validation | Schematic PDF, BoM spreadsheet |
| 2 - PCB Layout | Routing, DRC KiCad Files | PCB fabrication order |
| 3 - Prototyping | Breadboard / protoboard assembly and initial power-on | Powered prototype, voltage checks |
| 4 - Firmware | Development per module (CC1101, NFC, RFID, SD) | Firmware repo, unit tests |
| 5 - Integration | System integration, module interaction, OLED UI | Working integrated prototype |
| 6 - Validation | Functional test against acceptance criteria | Test report |
| 7 - Documentation | CDC finalisation, code documentation, README | Final CDC PDF, GitHub |

# 7. Acceptance Criteria & Test Plan

## 7.1 Functional Tests

| Test ID | Module | Procedure | Expected Result | Pass Criteria |
|---|---|---|---|---|
| T-01 | Power | Apply 5 V USB; measure 5 V and 3.3 V rails | Both rails within ±5% | Voltmeter reading in spec |
| T-02 | MCU | Flash blink firmware; observe LED | LED blinks at 1 Hz | Visible blink |
| T-03 | OLED | Flash display test; observe screen | All pixels lit, menu displayed | Visual Confirmation |
| T-04 | SD Card | Write and read test file via firmware | File written and read back correctly | Data integrity 100% |
| T-05 | CC1101 | Capture 433 MHz remote signal; display capture | Signal captured and stored on SD | Valid file on SD card |

| | | | | |
|---|---|---|---|---|
| T-06 | CC1101 | Replay stored 433 MHz signal to target device | Target device responds | Repeatable successful replay |
| T-07 | NFC | Read ISO 14443-A tag; display UID on OLED | UID displayed correctly | Matches reference reader |
| T-08 | RFID | Read EM4100 125 kHz tag; display data | Tag data displayed correctly | Matches reference reader |
| T-09 | Battery | Operate 4 h on battery; monitor voltage | Device operational for ≥ 4 h | No shutdown under load |
| T-10 | Buttons | Press each button; verify OLED response | Correct navigation action per button | All 6 buttons functional |

## 7.2 Compliance Checks

- All RF transmissions verified to comply with 433 MHz ISM band limits
- Device shall not transmit on unauthorised frequencies

# 8. References & Datasheets

| Document | Source |
|---|---|
| ATmega2560 Datasheet | https://www.microchip.com/en-us/product/atmega2560 |
| CC1101 Datasheet | todo |
| ST25R3916B Datasheet | https://www.st.com/resource/en/datasheet/st25r3916b.pdf |
| EM4095 Datasheet | https://www.emmicroelectronic.com/sites/default/files/products/datasheets/em4095_ds.pdf |
| microSD Socket (Molex) | Part 1040310811 |
| Project GitHub Repository | https://github.com/youpaw/elec-module |
| Flipper Zero Reference | https://flipperzero.one |

# 9. Notes

## 9.1 Hardware Block Diagram Description

The uploaded block diagram (firstDiagram.pdf) shows the following interconnections for the FLIPPER 42 system.